

Morris Alper, Ph.D.

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I research creative multimodal machine learning inspired by linguistics—how structures encoded by language provide a tool for interpreting and controlling modalities such as vision and audio, ML as a creativity aid, and innovative methods for low-resource languages.

Education

- Ph.D., Tel Aviv University** 2023 – 2025
Electrical Engineering (Machine Learning)
Advisors: *Dr. Hadar Averbuch-Elor, Dr. Raja Giryes*
Dissertation: *Multimodal Learning for High-Level Semantic Understanding: From Interpretability to Real-World Applications*
- M.Sc. (cum laude), Tel Aviv University** 2019 – 2023
Computer Science
Advisor: *Dr. Hadar Averbuch-Elor*
Thesis: *Learning Human-Human Interactions in Images with Weak Textual Supervision*
GPA: 98 / 100
- B.Sc., Massachusetts Institute of Technology (MIT)** 2012 – 2016
Linguistics and Mathematics (dual major). GPA: 4.9 / 5

Employment History

- Research Analyst (Postdoctoral), Carnegie Mellon University** 2025 – ongoing
Advisor: *Dr. Lori Levin*
Developing machine learning methods for endangered language documentation.
- Research Scientist Intern, Meta** 2024
Developing 3D generative models using weakly-structured image collections.
- Data Science Lead and Lecturer, Israel Tech Challenge** 2019 – 2023
Designed syllabus, wrote, and delivered lectures on data science to industry professionals.
- NLP Data Scientist, Pipl, Inc.** 2019 – 2021
Designed information extraction algorithms for unstructured textual web data.
- NLP Data Scientist, Reverso-Softissimo** 2018 – 2019
Designed NLP algorithms to automatically correct and enrich linguistic data.
- Software Developer, K Dictionaries** 2017 – 2018
Managed API for lexicographic data, and developed algorithms for data enrichment.
- Algorithm Development Intern, SECOM** 2016
Developed algorithms for improved facial recognition.
- Software and Algorithm Developer, Voiceitt** 2015 – 2016
Prototyped speech recognition algorithms and integrated them into the company's API.
- Undergraduate Research in Cognitive Linguistics, TedLab, MIT** 2012–2015
Analyzed psycholinguistic survey data to study cognition and language.

Research Publications

* indicates equal contribution.

Peer-Reviewed Conference Papers

1. Morris Alper*, Moran Yanuka*, Raja Giryes, Gašper Beguš. ConlangCrafter: Constructing Languages with a Multi-Hop LLM Pipeline. *ACL*, 2026. (*Oral*)
2. Morris Alper, David Novotny, Filippas Kokkinos, Hadar Averbuch-Elor, Tom Monnier. WildCAT3D: Appearance-Aware Multi-View Diffusion in the Wild. *NeurIPS*, 2025.
3. Rachel Mikulinsky*, Morris Alper*, Shai Gordin, Enrique Jiménez, Yoram Cohen, Hadar Averbuch-Elor. ProtoSnap: Prototype Alignment For Cuneiform Signs. *ICLR*, 2025.
4. Keren Ganon*, Morris Alper*, Rachel Mikulinsky, Hadar Averbuch-Elor. WAFFLE: Multimodal Floorplan Understanding in the Wild. *WACV*, 2025. (*Oral*)
5. Morris Alper and Hadar Averbuch-Elor. Emergent Visual-Semantic Hierarchies in Image-Text Representations. *ECCV*, 2024. (*Oral*)
6. Assaf Ben-Kish, Moran Yanuka, Morris Alper, Raja Giryes, Hadar Averbuch-Elor. Mitigating Open-Vocabulary Caption Hallucinations. *EMNLP*, 2024.
7. Moran Yanuka, Morris Alper, Hadar Averbuch-Elor, Raja Giryes. ICC: Quantifying Image Caption Concreteness for Multimodal Dataset Curation. *ACL (Findings)*, 2024.
8. Chen Dudai*, Morris Alper*, Hana Bezalel, Rana Hanocka, Itai Lang, and Hadar Averbuch-Elor. HaLo-NeRF: Learning Geometry-Guided Semantics for Exploring Unconstrained Photo Collections. *EG*, 2024.
9. Morris Alper and Hadar Averbuch-Elor. Kiki or Bouba? Sound Symbolism in Vision-and-Language Models. *NeurIPS*, 2023. (*Spotlight*)
10. Morris Alper and Hadar Averbuch-Elor. Learning Human-Human Interactions in Images from Weak Textual Supervision. *ICCV*, 2023.
11. Morris Alper*, Michael Fiman*, and Hadar Averbuch-Elor. Is BERT Blind? Exploring the Effect of Vision-and-Language Pretraining on Visual Language Understanding. *CVPR*, 2023.

Workshop and Non-Peer-Reviewed Papers

1. Morris Alper, Samopriya Basu, Nathan Bennett, Ryan Kessler, S Verlaine Ravana, Wendy F. K'ah Skáahluwaa Todd. Towards Low-Resource Text-to-Speech Generation for Indigenous Pacific Northwest Languages: The Case of Haida Xaad Kil. *ICSNL*, 2025.
2. Shai Gordin, Morris Alper, Avital Romach, Luis Saenz Santos, Naama Yochai, Roey Lalazar. CuReD: Deep Learning OCR for Cuneiform Text Editions and Legacy Materials. *ML4AL Workshop*, 2024.
3. Morris Alper. Auto-generating Bilingual Dictionaries: Results of the TIAD-2017 Shared Task Baseline Algorithm. *LDK Workshops*, 2017.

Preprints and Under Review

1. Morris Alper*, Enora Rice*, Bhargav Shaldilya*, Alexis Palmer, Lori Levin. CWoMP: Morpheme Representation Learning for Interlinear Glossing. 2026
2. Yakov Kolani, Maxim Melichov, Cobi Calev, Morris Alper. Phonikud: Hebrew Grapheme-to-Phoneme Conversion for Real-Time Text-to-Speech. 2025.
3. Shahaf Pruss*, Morris Alper*, Hadar Averbuch-Elor. Dynamic Scene Understanding from Vision-Language Representations. 2025.

Teaching Experience

Human Language for Artificial Intelligence, teaching assistant, Carnegie Mellon University Fall 2025

Digital Signal Processing, teaching assistant, Tel Aviv University Fall 2023

Data Science Lead and Lecturer, Israel Tech Challenge 2019 – 2023

- Designed syllabus spanning fundamentals through advanced topics (deep learning, NLP, computer vision)

- Lectured on data science and machine learning to diverse industry professionals across multiple cohorts
- Mentored student capstone projects in technical implementation and communicating results
- Achieved strong student evaluations (consistently above 6.0 on 7-point scale)

Grants and Funding

Awarded

Paleography Powered by Machine Learning: Analyzing Cuneiform Documents with Neural Networks 2023

LMU-TAU Research Cooperation Program

Role: Grant co-writer and funded researcher (PIs: Yoram Cohen, Hadar Averbuch-Elor, Enrique Jiménez)

Pending

Tracing the Signs: Multimodal Foundation Models for Ancient Writing 2026

Level I Grant, Humanities and Artificial Intelligence Virtual Institute (HAVI; Schmidt Sciences)

Role: Co-PI (Lead PI: Ahmad Al-Jallad)

Reading the Rocks: AI-powered Digital Editions and the Preservation of Endangered pre-Islamic Arabian Epigraphic Heritage 2025

Implementation Grant, Scholarly Editions and Scholarly Translation (NEH-BSF)

Role: Senior Computational Specialist (Lead PI: Ahmad Al-Jallad)

Talks and Presentations

Invited Talks (Selected)

- 2026 **CWoMP: Morpheme Representation Learning for Interlinear Glossing.** NLP Seminar, National Research Council Canada.
- 2026 **Phonikud: Overcoming Phonetic Underspecification for Hebrew Text-To-Speech.** WAVLab Seminar, CMU.
- 2026 **Old Words, New Words: ML for Ancient and Fictional Languages.** Guest Lecture for Linguistics 265: Biological and Artificial Language, UC Berkeley.
- 2025 **Old Words, New Words: ML for Ancient and Fictional Languages.** IBM Israel.
- 2025 **Understanding the Knowledge of Vision-Language Models.** SC Lab Seminar, UC Berkeley.
- 2025 **Emergent Visual-Semantic Hierarchies in Image-Text Representations.** NYC Computer Vision Day 2025.
- 2024 **Understanding the Knowledge of Vision-Language Models.** IMAGINE Lab Seminar, Ecole Nationale des Ponts et Chaussées.
- 2024 **Emergent Visual-Semantic Hierarchies in Image-Text Representations.** Vision & AI Seminar, Weizmann Institute of Science.
- 2024 **Kiki or Bouba? Sound Symbolism in Vision-and-Language Models.** Israel Machine Vision Conference (IMVC).
- 2023 **What is Learned by Vision and Language Models.** School of Electrical and Computer Engineering, Ben Gurion University.
- 2023 **What is Learned by Vision and Language Models.** Shevah Mofet High School Linguistics Club.
- 2023 **Is BERT Blind? Exploring the Effect of Vision-and-Language Pretraining on Visual Language Understanding.** Vision Seminar, Hebrew University of Jerusalem.
- 2023 **“Diplomatics”: Deep learning for Automatic Analysis of Cuneiform Texts.** First Meeting of the Digital Ancient Near Eastern Studies Network (DANES).
- 2022 **“Diplomatics”: Deep learning for Automatic Analysis of Cuneiform Texts.** LMU-TAU Workshop for Early-Career Researchers.
- 2017 **Auto generating bilingual dictionaries.** eLex 2017 conference, Leiden, the Netherlands.

Industry Workshops (delivered through Israel Tech Challenge)

2021–2022 **NLP Workshop Series**. AT&T Israel.

2020 **Cutting-Edge NLP With Transformers and BERT**. PayPal Israel.

Recorded Presentations and Interviews

2024 IMVC 2024 presentation on “Kiki or Bouba?” NeurIPS 2023 paper [video]

2024 Radio interview for Kan Tarbut program on VLM sound symbolism research (46:30-60:00, Hebrew language) [audio]

2023 “Deeplomatics” presentation at DANES [video]

2017 “Auto generating bilingual dictionaries” at eLex [video]

Miscellaneous Experience

Awards and Achievements

2011 **International Linguistics Olympiad**, **first place internationally** (both individual and team rounds), best solution (problem 2), Zhurinsky Memorial Prize.

2009 **International Linguistics Olympiad**, honorable mention (individual round), first place in team round.

2008 **International Linguistics Olympiad**, tied for 6th place individual round (silver medal).

Peer Review Service

2026 ACL ARR (Jan, Mar), (Ongoing) CoLM

2025 SIGGRAPH, NeurIPS (*Top Reviewer*), ICLR, ACL ARR

2024 COLM, NeurIPS, ACML, ICLR

Languages

English (native), **Hebrew** (professional), **French** (upper intermediate)